

The State of Privacy Policy Regulatory Compliance for Educational Technology

Full Paper

Abstract

Educational technology (EdTech) delivers meaningful learning benefits but also introduces significant privacy and regulatory compliance risks. Regulations such as the GDPR and COPPA require vendors to publish privacy policies disclosing what personal data are collected, how they are processed, and with whom they are shared, establishing formal governance mechanisms intended to support organizational oversight and privacy risk management. Children are particularly vulnerable because adoption and data-disclosure decisions are typically made on their behalf by schools and administrators. To evaluate the effectiveness of privacy policies as a compliance mechanism, we scrape and analyze the policies of a random sample of 1,500 EdTech applications used across U.S. school districts and assess their adherence to key statutory disclosure requirements. Our findings show that many deployed applications fail to meet these standards, revealing systematic gaps between formal regulatory obligations and actual disclosure practices. We interpret these results using institutional and signaling theory, highlighting how weak enforcement and symbolic compliance shape vendor behavior in regulated digital ecosystems.

Keywords

EdTech, information privacy, privacy policies, regulatory compliance, institutional theory

Introduction

Educational technology (EdTech) delivers meaningful learning benefits across educational contexts but also introduces substantial privacy and regulatory compliance risks. Children in school represent a particularly vulnerable population (Nissenbaum, 2004) because technology adoption and data-disclosure decisions are typically made on their behalf by schools, teachers, and administrators rather than by the children themselves (Walker et al., 2022). Regulations such as the General Data Protection Regulation (GDPR) and the Children’s Online Privacy Protection Act (COPPA) therefore require vendors to provide clear public disclosures describing what personal data are collected, how those data are processed, and with whom they are shared (GDPR, 2016). While privacy policies can function as governance artifacts and compliance signals, their existence does not ensure that vendors’ operational practices match their disclosures or that decision makers can reliably infer data practices from policy language alone (Reidenberg et al., 2015). These regulations thus establish minimum disclosure expectations intended to support transparency, oversight, and privacy risk management in organizational technology adoption decisions.

This governance challenge is practically consequential because children’s data are routinely exposed to third parties through products used for schooling. Large-scale testing of apps used in U.S. K–12 schools found that most transmitted student data to third parties, frequently including advertising and analytics entities (Internet Safety Labs, 2022). Independent investigations of online learning tools adopted during the COVID-19 period likewise documented widespread tracking technologies capable of monitoring children beyond the educational context (Human Rights Watch, 2022). These risks are amplified by the structure of school technology adoption. Decisions are typically made by teachers and administrators who lack the technical capacity to independently verify data flows through network or code inspection, leaving privacy policies and similar disclosures as the primary governance signals available at scale (Chanenson, 2023). At the same time, tool sprawl in K–12 districts is substantial, districts may access thousands of digital tools annually, making comprehensive technical vetting infeasible and increasing reliance on vendor disclosures (Merod, 2023). Frequent policy updates further complicate oversight, suggesting that policy-based governance requires ongoing evaluation rather than one-time review (Lightspeed Systems, 2022).

These practical realities motivate the central research question of this study: *How well do current EdTech vendor privacy policies adhere to relevant regulatory requirements?* Addressing this question requires methodological choices that reflect how privacy information is actually encountered and used by school decision makers. A growing stream of privacy research has relied on platform-provided “labels” or short-form disclosures (e.g., Google Play’s Data Safety section or Apple’s App Privacy “nutrition labels”) as the unit of analysis. However, multiple studies and audits report discrepancies between these label disclosures and the information contained in linked privacy policies or other disclosures, raising questions about whether store-level statements are sufficiently complete or reliable for compliance assessment (Mozilla Foundation, 2023). In the EdTech context, this limitation is important because district decision makers frequently rely on vendor websites, district procurement workflows, or linked policy documents—sources that may contain more detailed, legally oriented disclosures than platform summaries. Moreover, sampling strategies based on “top N” apps in public app stores may underrepresent the specific ecosystem of tools actually deployed in U.S. schools, where district adoption is shaped by procurement, interoperability, and institutional constraints rather than consumer popularity alone (Internet Safety Labs, 2022). To better match the study context, we assemble a random sample of local education agencies (LEAs) across all 50 states and identify apps in use by manually reviewing district websites for tools recommended or required for students. We then scrape and analyze the corresponding vendor privacy policies.

Understanding why disclosure compliance varies across vendors requires theoretical explanation beyond descriptive measurement. Theoretically, privacy-policy compliance in EdTech presents a governance puzzle that has been articulated empirically but is less frequently explained through IS theory. Much of the existing evidence on children’s digital privacy and app tracking comes from computer science and policy communities, where the primary objective is often measurement rather than theory building (e.g., Chanenson, 2023). We therefore draw on complementary theoretical lenses to interpret why compliance may be uneven even when regulatory expectations are well specified. Institutional theory highlights how coercive, normative, and mimetic pressures can produce both substantive compliance and symbolic conformity, particularly where enforcement is weak or fragmented (DiMaggio & Powell, 1983). Signaling theory explains how privacy policies may function as market-facing signals intended to reduce perceived risk or satisfy procurement checklists, even when the signal is low cost or weakly verified (Spence, 1973; Pavlou et al, 2007). Contextual integrity foregrounds why data flows that extend beyond the educational context may be perceived as inappropriate even when disclosed, emphasizing the importance of context-relative norms in evaluating “acceptable” information practices (Nissenbaum, 2004).

To examine the effectiveness of privacy policies as a governance and compliance mechanism, we scrape and analyze the privacy policies of a random sample of 1,500 EdTech applications used across U.S. school districts. We assess their adherence to key statutory disclosure requirements under GDPR and COPPA. Our findings indicate that a substantial proportion of EdTech apps currently deployed in U.S. schools fail to meet these regulatory standards, revealing systematic gaps between formal compliance obligations and actual disclosure practices. We interpret these results through the lenses of institutional theory and signaling theory, highlighting how weak enforcement, symbolic compliance, and market signaling incentives shape vendor behavior in regulated digital ecosystems. The study contributes to research on information security governance and privacy compliance by empirically evaluating privacy policies as an institutional control mechanism in a high-risk, child-centered domain.

Regulatory Requirements for Privacy Policies

What exactly must EdTech vendors disclose in their privacy policies to meet regulatory expectations? A comprehensive review of all applicable federal, state, and international student privacy regulations is beyond the scope of this study. However, two of the most influential and widely cited regulatory frameworks governing digital privacy disclosures are GDPR (European Union, 2016) and COPPA (Federal Trade Commission, 2013). Both regulations establish explicit transparency requirements that organizations must satisfy through publicly available privacy policies or equivalent disclosures. These policies function as governance artifacts intended to inform organizational decision makers about data practices and to support oversight, risk assessment, and compliance during technology adoption.

GDPR and COPPA specify a set of required disclosures concerning the collection, use, sharing, and governance of personal data. At a high level, both regulations require vendors to identify the entity responsible for data practices, describe the categories of personal information collected, explain the purposes for which data are used, and disclose whether and with whom data are shared. GDPR further requires disclosure of the legal basis for processing, data retention practices, international data transfers, and data subject rights, including rights to access, deletion, and portability. COPPA emphasizes transparency regarding parental rights, consent procedures, third-party data sharing, and security practices when children's data are involved. Together, these provisions define a baseline set of disclosures intended to enable informed decision making and regulatory accountability.

To operationalize these requirements, we synthesized the disclosure expectations specified in GDPR and COPPA into a unified compliance framework. Table 1 summarizes the specific elements that privacy policies must include to meet core regulatory transparency obligations under these statutes. This framework serves as the basis for our coding and analysis. For each EdTech vendor privacy policy in our sample, we evaluate whether required disclosure elements are present, enabling systematic assessment of regulatory adherence across vendors and applications.

Requirement category	Specific requirement (must be disclosed in privacy policy)	GDPR (citation)	COPPA (citation)
Company identity	Name and contact details of data controller/operator	Art. 13(1)(a), 14(1)(a)	§312.4(d)(1)
Company identity	Contact details of Data Protection Officer (if applicable)	Art. 13(1)(b), 14(1)(b)	—
Company identity	Name, address, email, and contact of all operators collecting children's data	—	§312.4(d)(1)
Types of data collected	Categories of personal data collected	Art. 13(1)(c), 14(1)(d)	—
Types of data collected	Types of personal information collected from children	—	§312.4(d)(2)
Types of data collected	Collection of persistent identifiers (cookies, device IDs) if used	—	§312.2; §312.4(d)(2)
Purpose of collection/use	Purposes of processing personal data	Art. 13(1)(c)	—
Purpose of collection/use	Legal basis for processing (consent, contract, etc.)	Art. 6; Art. 13(1)(c)	—
Purpose of collection/use	How children's information is used	—	§312.4(d)(2)
Third-party sharing	Recipients or categories of recipients of personal data	Art. 13(1)(e), 14(1)(e)	—

Third-party sharing	Third parties collecting or receiving children's data	—	§312.4(d)(3)
Third-party sharing	Purpose of third-party disclosure	—	§312.4(d)(3)
International transfers	Transfers outside EU and safeguards used	Art. 13(1)(f)	—
Retention	Data retention period or criteria used to determine retention	Art. 13(2)(a)	—
Retention	Statement that children's data retained only as long as necessary	—	§312.10
User/parent rights	Right to access personal data	Art. 15	—
User/parent rights	Right to rectification	Art. 16	—
User/parent rights	Right to erasure ("right to be forgotten")	Art. 17	—
User/parent rights	Right to restrict processing	Art. 18	—
User/parent rights	Right to data portability	Art. 20	—
User/parent rights	Right to object to processing	Art. 21	—
User/parent rights	Right to withdraw consent	Art. 13(2)(c)	—
User/parent rights	Right to lodge complaint with supervisory authority	Art. 13(2)(d)	—
User/parent rights	Parent right to review child data	—	§312.6
User/parent rights	Parent right to delete child data	—	§312.6(a)(2)
User/parent rights	Parent right to refuse further collection/use	—	§312.6(a)(1)
Automated decision-making	Disclosure of profiling or automated decision-making	Art. 13(2)(f); Art. 22	—
Data provision requirements	Whether data provision is mandatory and consequences of refusal	Art. 13(2)(e)	—
Security	Description of reasonable security safeguards	—	§312.8

Security	Appropriate technical/organizational security measures (transparency expectation)	Art. 5(1)(f); Art. 32	—
Policy accessibility	Information must be concise, transparent, intelligible	Art. 12(1)	—
Policy accessibility	Clear and prominent privacy policy link online	—	§312.4(d)
Updates/changes	Notice of material changes to practices	—	§312.4(b); §312.5(c)
Consent mechanisms	Description of parental consent procedures	—	§312.5
Data source	Source of data if not collected directly from user	Art. 14(2)(f)	—

Table 1. Regulatory Disclosure Requirements for EdTech Privacy Policies

Prior Research on Privacy Policies and Children’s Digital Privacy

A substantial body of research has examined the transparency, readability, and completeness of online privacy policies. Early work demonstrated that privacy policies are often difficult for users to interpret and may not effectively communicate data practices, limiting their usefulness as decision-making tools (McDonald & Cranor, 2008; Reidenberg et al., 2015). Subsequent studies have documented inconsistencies between stated policies and actual data practices, as well as internal contradictions that undermine the reliability of notice-and-consent frameworks (Okoyomon et al., 2019).

Large-scale computational analyses have further evaluated the extent to which mobile application privacy policies satisfy regulatory disclosure expectations. For example, Zimmeck et al. (2019) developed automated methods for assessing policy compliance at scale, while more recent work has evaluated the completeness of privacy policies relative to GDPR requirements across large samples of mobile applications (Xiang et al., 2023; Alamri et al., 2022; Liu et al., 2022). Collectively, this research shows that privacy policies frequently omit required disclosures or provide incomplete information, raising concerns about their effectiveness as transparency and compliance mechanisms.

A parallel stream of research has focused specifically on children’s digital privacy and regulatory compliance. Empirical studies have shown that many child-directed applications collect and transmit personal data in ways that may conflict with stated policies or regulatory expectations (Reyes et al., 2018). More recent work has continued to document extensive data collection and tracking practices in apps used by or directed toward children, highlighting persistent gaps between regulatory requirements and actual data practices (Cecere et al., 2025). This literature underscores the heightened risks associated with digital services used by minors and the importance of effective governance mechanisms for evaluating vendor data practices.

Recent research has also examined platform-provided privacy disclosures, such as app store privacy labels and summaries. These studies find that platform-level disclosures may be incomplete, inconsistent, or misaligned with the more detailed privacy policies linked to applications, raising questions about their reliability for assessing regulatory compliance or informing adoption decisions (Mozilla Foundation, 2023). While such disclosures provide useful signals, they may not capture the full scope of information required under applicable privacy regulations.

Despite growing attention to privacy policies and children's digital privacy, several important gaps remain. First, much of the existing literature examines general consumer applications or children's apps broadly rather than the specific ecosystem of educational technologies used in K–12 school settings. Second, prior studies frequently rely on app store disclosures, network-traffic analysis, or small convenience samples rather than systematically evaluating full vendor privacy policies against formal regulatory disclosure requirements. Third, sampling approaches often focus on popular or highly ranked applications rather than those actually deployed in schools, limiting contextual validity for educational decision making. Finally, much of the existing work is descriptive or technical in orientation and offers limited theoretical explanation for why regulatory disclosure compliance varies across vendors.

This study addresses these gaps by evaluating the privacy policies of a large sample of EdTech applications actually used across U.S. school districts, assessing their adherence to core disclosure requirements derived from GDPR and COPPA, and interpreting observed compliance patterns through established IS theoretical lenses.

In addition, many large-scale privacy policy and compliance studies rely primarily on datasets drawn from the Google Play store due to the relative ease of large-scale Android app scraping and analysis. While methodologically practical, this approach may not fully represent the ecosystem of applications used in educational settings, where both iOS and Android platforms are widely deployed. By incorporating applications available across platforms and focusing specifically on tools adopted by school districts, this study provides a more contextually representative assessment of privacy policy compliance in the EdTech environment.

Theoretical Explanation of Vendor Behaviors

Variation in privacy policy completeness across EdTech vendors requires explanation beyond descriptive measurement. Although regulations such as GDPR and COPPA specify required disclosures, organizations differ substantially in how fully these expectations are reflected in publicly available policies. To explain this variation, we draw primarily on institutional theory, complemented by signaling theory. Prior research has applied these perspectives to organizational privacy and security practices broadly, but rarely to the completeness of vendor privacy disclosures in the educational technology sector.

Institutional theory suggests that organizational practices are shaped by coercive, normative, and mimetic pressures within an organizational field (DiMaggio & Powell, 1983). Regulatory requirements create coercive pressures for organizations to adopt formal disclosure practices, while professional norms and industry expectations generate additional incentives to demonstrate responsible data governance. However, organizations frequently respond to such pressures symbolically rather than substantively, adopting formal structures that signal compliance without fully implementing underlying requirements (Edelman, 1992). Institutional perspectives have been widely used to explain organizational responses to privacy and information security requirements, including the adoption of formal governance mechanisms that vary in depth and effectiveness (Bélanger & Crossler, 2011).

These dynamics are likely to produce uneven privacy policy compliance across EdTech vendors. Larger and more established firms face greater scrutiny from regulators, institutional customers, and the public, increasing incentives to maintain more complete and defensible privacy disclosures. Smaller or newer vendors may face weaker institutional pressures and fewer enforcement risks, resulting in less comprehensive policies. Institutional theory therefore suggests that compliance will be uneven across vendors and that some degree of symbolic compliance, where policies exist but omit required disclosures, will be common. We thus expect to observe mixed levels of regulatory disclosure completeness across EdTech vendor privacy policies, with more complete disclosures among larger and more visible vendors.

Signaling theory provides a complementary explanation for why vendors publish privacy policies and how those policies function in technology adoption decisions (Spence, 1973; Connelly et al., 2011). In markets characterized by information asymmetry, organizations use observable signals to convey unobservable qualities such as trustworthiness, legitimacy, or regulatory compliance. Prior research in information systems has shown that firms use publicly available disclosures and governance mechanisms to reduce uncertainty and build trust in digital exchange environments (Pavlou & Gefen, 2004; Pavlou et al., 2007).

Privacy policies can serve as such signals, particularly in educational settings where schools and districts must evaluate vendor risk but often lack the technical capacity to independently verify data practices. Vendors therefore have incentives to publish policies that signal compliance and responsibility to potential adopters, even when the completeness of those disclosures varies. Larger vendors with greater reputational exposure may invest more heavily in producing comprehensive signals, while smaller vendors may provide minimal disclosures sufficient to maintain market participation. From a signaling perspective, we therefore expect most vendors to provide at least some form of privacy policy disclosure, but with substantial variation in completeness and depth, reflecting differences in signaling incentives and resource investment.

Together, institutional and signaling perspectives suggest that privacy policies will function as imperfect governance signals in the EdTech marketplace: widely present but uneven in their alignment with regulatory disclosure requirements.

Methodology

Research design and sample construction

This study evaluates the extent to which EdTech vendor privacy policies adhere to regulatory disclosure requirements derived from GDPR and COPPA. To ensure contextual validity, we constructed a sample of applications actually used in U.S. K–12 school environments rather than relying on convenience samples drawn from app store rankings.

We generated a random sample of local education agencies (LEAs) across all 50 U.S. states and manually reviewed district websites to identify educational technology applications recommended or required for student use. This approach captures tools deployed in real instructional settings rather than those selected solely on the basis of popularity within commercial app marketplaces. For each identified application, we located and archived the corresponding vendor privacy policy from vendor websites or linked disclosures.

The resulting dataset includes approximately 1,500 EdTech applications used across diverse school districts and grade levels. The sample includes applications available across both Apple iOS and Android ecosystems, providing broader coverage of the EdTech environment than studies relying on single-platform app store datasets.

Measures

To evaluate regulatory adherence, we developed a disclosure-based compliance framework derived from transparency requirements specified in the General Data Protection Regulation (GDPR) and the Children’s Online Privacy Protection Act (COPPA). These regulations establish explicit expectations regarding disclosure of data collection, use, sharing, retention, and user rights.

The disclosure elements summarized in Table 1 were synthesized into a unified coding instrument. Each requirement was operationalized as a binary indicator reflecting whether the corresponding disclosure element was present in a vendor’s publicly available privacy policy. This approach enables systematic assessment of whether policies include the types of information required under core regulatory transparency expectations.

Because this study evaluates publicly available disclosures rather than legal determinations of compliance, the analysis focuses on the presence or absence of required policy elements rather than the legal sufficiency of specific language.

Given the scale and textual complexity of privacy policies, we employed ChatGPT 5.2 to assist in identifying and classifying disclosure elements. Earlier LLM models have been demonstrated to have particularly promising results for evaluating privacy policies (Rodriguez et al., 2024; Tang et al., 2023). Structured prompts were used to evaluate whether specific disclosure requirements derived from Table 1

were present in each policy, generating binary indicators for each disclosure category. Prompts were designed to identify clearly observable disclosure elements (e.g., retention statements, third-party sharing disclosures, identification of data controllers) rather than subjective assessments of policy quality. Policies were processed using standardized prompts and procedures to ensure consistency across vendors.

The primary outcome of interest is the completeness of regulatory disclosure within vendor privacy policies. For each policy, we computed descriptive measures reflecting the presence of required disclosure elements across GDPR- and COPPA-derived categories. These measures enable assessment of overall compliance patterns and identification of commonly omitted disclosures.

To explore expectations derived from institutional and signaling perspectives, we constructed proxy indicators of vendor visibility and scale using observable platform metrics such as number of downloads and user review counts. These indicators provide approximate measures of organizational prominence and market exposure, which theory suggests may influence disclosure completeness.

Results

Discussion

REFERENCES

- Internet Safety Labs, (2022), Internet Safety Labs Research Reveals 96% of School Apps Send Student Data to Third Parties, Including Advertisers, Retrieved Feb 4, 2026 from <https://internetsafetylabs.org/blog/news-press/isl-research-reveals-96-of-school-apps-send-student-data-to-third-parties>
- Human Rights Watch, (2022), “How Dare They Peep into My Private Life?”, Retrieved Feb 4, 2026 from <https://www.hrw.org/report/2022/05/25/how-dare-they-peep-my-private-life/childrens-rights-violations-governments>
- Chanenson, J., Sloane, B., Rajan, N., Morril, A., Chee, J., Huang, D. Y., & Chetty, M. (2023, April). Uncovering privacy and security challenges in K-12 schools. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (pp. 1-28).
- Merod, Anna. (2023), Districts used 2,591 ed tech tools on average in 2022-23, Retrieved Feb 4, 2026 from <https://www.k12dive.com/news/school-districts-ed-tech-use/685995/>
- Lightspeed Systems, (2022), 2022 Edtech App Report, Retrieved Feb 4, 2026 from <https://www.lightspeedsystems.com/ebook/edtech-app-report/>
- Mozilla Foundation, (2023), See No Evil: Loopholes in Google’s Data Safety Labels Keep Companies in the Clear and Consumers in the Dark, Retrieved Feb 4, 2026 from <https://www.mozillafoundation.org/nl/campaigns/googles-data-safety-labels/>
- Walker, K. L., Bodendorf, K., Kiesler, T., de Mattos, G., Rostom, M., & Elkordy, A. (2023). Compulsory technology adoption and adaptation in education: A looming student privacy problem. *Journal of Consumer Affairs*, 57(1), 445-478.
- GDPR, (2016), Article 13 GDPR: Information to be provided where personal data are collected from the data subject, Retrieved Feb 4, 2026 <https://gdpr-info.eu/art-13-gdpr/>
- Reidenberg, J. R., Breaux, T. D., Cranor, L. F., French, B., Grannis, A., Graves, J. T., Liu, F., McDonald, A. M., Norton, T. B., Ramanath, R., Schaub, F., & Wilson, N. (2015). Disagreeable privacy policies: Mismatches between meaning and users’ understanding. *Berkeley Technology Law Journal*, 30(1), 39–88. https://ir.lawnet.fordham.edu/cgi/viewcontent.cgi?article=1618&context=faculty_scholarship

- DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2), 147–160.
- Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3), 355–374.
- Pavlou, P. A., Liang, H., & Xue, Y. (2007). Understanding and mitigating uncertainty in online exchange relationships. *MIS Quarterly*, 31(1), 105–136.
- Nissenbaum, H. (2004). Privacy as contextual integrity. *Washington Law Review*, 79(1), 119–158.
- European Union. (2016). *Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation)*.
- Federal Trade Commission. (2013). *Children's Online Privacy Protection Rule, 16 C.F.R. Part 312*. <https://www.ecfr.gov/current/title-16/chapter-I/subchapter-C/part-312>
- McDonald, A. M., & Cranor, L. F. (2008). The cost of reading privacy policies. *I/S: A Journal of Law and Policy for the Information Society*, 4(3), 543–568.
- Reidenberg, J. R., Breaux, T. D., Cranor, L. F., French, B., Grannis, A., Graves, J. T., Liu, F., McDonald, A. M., Norton, T. B., Ramanath, R., Schaub, F., & Wilson, N. (2015). Disagreeable privacy policies: Mismatches between meaning and users' understanding. *Berkeley Technology Law Journal*, 30(1), 39–88.
- Okoyomon, E., Samarin, N., Wijesekera, P., Elazari Bar On, A., Vallina-Rodriguez, N., Reyes, I., Feal, Á., & Egelman, S. (2019). On the ridiculousness of notice and consent: Contradictions in app privacy policies. In *Proceedings on Privacy Enhancing Technologies (PoPETs)* (Vol. 2019, Issue 3, pp. 173–193).
- Zimmeck, S., Story, P., Smullen, D., Ravichander, A., Wang, Z., Reidenberg, J. R., Russell, N. C., & Sadeh, N. (2019). MAPS: Scaling privacy compliance analysis to a million apps. In *Proceedings on Privacy Enhancing Technologies (PoPETs)* (Vol. 2019, Issue 3, pp. 66–86).
- Xiang, A., Pei, W., & Yue, C. (2023). PolicyChecker: Analyzing the GDPR completeness of mobile apps' privacy policies. In *Proceedings of the ACM Web Conference 2023*. Association for Computing Machinery.
- Alamri, H., Maple, C., Mohamad, S., & Epiphaniou, G. (2022). Do the right thing: A privacy policy adherence analysis of over two million apps in Apple iOS App Store. *Sensors*, 22(23), 9176.
- Liu, K., Xu, G., Zhang, X., Xu, G., & Zhao, Z. (2022). Evaluating the privacy policy of Android apps: A privacy policy compliance study for popular apps in China and Europe. *Security and Communication Networks*, 2022, Article 2508690.
- Reyes, I., Wijesekera, P., Reardon, J., Elazari Bar On, A., Razaghpanah, A., Vallina-Rodriguez, N., & Egelman, S. (2018). “Won’t somebody think of the children?” Examining COPPA compliance at scale. In *Proceedings on Privacy Enhancing Technologies (PoPETs)* (Vol. 2018, Issue 3, pp. 63–83).
- Cecere, G., Lefrere, V., Le Guel, F., Tucker, C., & Yin, P.-L. (2025). Privacy and platform governance: The case of apps for young children. *Academy of Management Perspectives*. Advance online publication.
- Bélanger, F., & Crossler, R. E. (2011). Privacy in the digital age: A review of information privacy research in information systems. *MIS Quarterly*, 35(4), 1017–1041.
- Connelly, B. L., Certo, S. T., Ireland, R. D., & Reutzel, C. R. (2011). Signaling theory: A review and assessment. *Journal of Management*, 37(1), 39–67. <https://doi.org/10.1177/0149206310388419>

Edelman, L. B. (1992). Legal ambiguity and symbolic structures: Organizational mediation of civil rights law. *American Journal of Sociology*, 97(6), 1531–1576.

Pavlou, P. A., & Gefen, D. (2004). Building effective online marketplaces with institution-based trust. *Information Systems Research*, 15(1), 37–59.

Rodriguez, D., Yang, I., Del Alamo, J. M., & Sadeh, N. (2024). Large language models: a new approach for privacy policy analysis at scale. *Computing*, 106(12), 3879–3903.

Tang, C., Liu, Z., Ma, C., Wu, Z., Li, Y., Liu, W., ... & Fan, L. (2023). PolicyGPT: Automated analysis of privacy policies with large language models. arXiv preprint arXiv:2309.10238.